

Student agency-as-achieved in a changing world with Agentic AI

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Introduction

Students are learning to work with Artificial Intelligence (AI) systems more and more (Kumar et al. 2024) as AI is playing a bigger part in student's academic work, school projects and school policy. Is this affecting their freedom to act?

The freedom to act according to one's own intention and volition is often described as **agency** (Skinner 1996). It is this freedom that is described as *achieved agency* when it is **realised** "through the active engagement of individuals with aspects of their contexts-for-action" (Biesta and Tedder 2007), that always exists in an temporal-relational *ecology* (Biesta and Tedder 2006). This *achieved agency* is under pressure when systems circulate responsibility among human and non-human actors (Leonardi 2025).

The distinction between human and technological agency blurs with Agentic AI (AAI), as AAI can operate (semi-)autonomously and pursue complex goals with little human intervention (Acharya, Kuppan, and Divya 2025). AAI uses instances of Generative AI (GenAI) interfacing with each other creating seemingly autonomous and agentic systems, as a next step from the GenAI chat-platforms widely in use among students already (Silvennoinen, Aksovaara, and Alanko-Turunen 2025).

The *iterational* aspect of Emirbayer and Mische (1998)'s agency upon which Biesta and Tedder (2007) based the *achieved agency* is exposed in structure and collective beliefs, as *truth* expressed in instruments such as policies or punitive action (Foucault 1980, 131), or as expressed by teachers in their narratives or dispositions (Nespor 1987). Beliefs about "inevitability of technology" and the cult of innovation (Winner 2018) pervade, while at the same time mistrust and weariness tries to keep AI out of universities (Guest et al. 2025). Technology is never just neutral nor is it inherently good or bad (Morrow 2014; Heyndels 2023) it is there to be assistive to our human needs (Arora 2024, 32), and changes us just as much as we change it (Latour 1994, 33).

Students navigate these punitive faculties, policies and teacher narratives as ecology in which they need to realize goals, their agency changing from situation to situation. Is Agentic AI

infringing on their intentional volitions realized as *achieved agency* as the temporal-relational *ecology* changes?

Knowledge Gap

Literature focusing on AAI is scarce, unlike the literature for GenAI, which is closely related as precursor to AAI. This shows the need to further examine the phenomenon of AAI in relation to students. Where GenAI is used in studies results can be used as corollary for the type of results for AAI.

The way AAI or GenAI is influencing agency in students seems not to have been studied, apart from the influence it has on critical thinking (Suriano et al. 2025), and why and how they use it for studying (Silvennoinen, Aksovaara, and Alanko-Turunen 2025).

While research has been done about future projections of human agency in conjunction with AI in decision making (Pew Research Center, Anderson, and Rainie 2023) and in education (Mouta, Pinto-Llorente, and Torrecilla-Sánchez 2025) it does not deal with how achieved agency for humans now. Or how AAI is changing realized intentions.

Theorizations of agency by Biesta and Tedder (2007) clarify that action flows from the values and beliefs of a person. This study will try to deepen understanding on this interplay of beliefs and agency realised in the context of Agentic AI among students. As action and belief are tied together closely (McCormick 2014) it is relevant to investigate these beliefs that inform action further.

Research Questions

How is Agentic AI changing values and beliefs of students in the ecology in which agency-as-achieved can be enacted?

Sub Questions

1. What values and beliefs are prevalent among students concerning Agentic AI?
2. How are prevalent beliefs disclosed in policy and teacher narratives relevant to students because of AAI?
3. In what ways are students performing academic and non-academic tasks, by either conforming or subverting rules and restrictions?

Methodology

Table 1: Research Questions and how they relate to methods

Question	Method	mini- mum num- ber of partic- ipants
RQ1 What values and beliefs are prevalent among students concerning Agentic AI?	Thematic Analysis / UMT	n = 6 / n ~ 200
RQ2 How are prevalent beliefs disclosed in policy and teacher narratives relevant to students because of AAI?	Critical Discourse Analysis	n = 5-8
RQ3 In what ways are students performing academic and non-academic tasks, by either conforming or subverting rules and restrictions?	Mixed-Method	n = 5-8 for inter-views n ~ 40 for self-report cards

RQ1: Thematic Analysis and UMT

Analysis

As clarified in the Knowledge Gap, there is a close relation between action and belief, however in the context of Agentic AI the values and beliefs among students. In Kramer and Engeström (2019) the word *worldview* is used to describe this belief, which in Biesta and Tedder (2006)’s definition would contribute to the *ecology*. The way this is investigated in Kramer and Engeström (2019) is through the use of Leontiev (2007)’s Ultimate Meanings Technique (UMT).

Where Kramer and Engeström (2019) used UMT to discover worldview among teachers I will investigate among a small group of students.

Thematic analysis will be used as Naeem et al. (2023) and Braun and Clarke (2021) describe it to find common themes and topics in personal qualitative interview with students concerning these worldviews.

After the initial analysis and recognized larger themes questions will be formulated to be able to use a more quantitative approach in the form of a survey.

RQ2: Critical Discourse Analysis

Through *critical discourse analysis* (CDA) I will look at what kind of language is constructing narratives, describes relations of power and how it creates or reinforces beliefs (Blommaert and Bulcaen 2000). With CDA I can take a look at school policy and try to find patterns in power dynamics and social effects in discourse. In further analysis I will analyze context and intertextuality to challenge social inequalities. I will focus mainly on step two: *Identify obstacles to addressing the social wrong* of the CDA process as described by Fairclough (2013).

1. Analyse dialectical relations
2. Select texts
3. Analyse texts interdiscursively and linguistically

To assist in the Critical Discourse analysis I will use the Cui et al. (2023) package for R to help quantify some aspects of it.

Besides investigating school policy, I will collect in-class field notes, transcripts of audio-recorded lectures and classroom talk, i.e. when students and teachers interact with each other in the classroom, and interviews with teachers.

To analyze this data I will codify the texts, look for synonyms and highlight recurring normative statements on the use of or the prevalence of AI technologies. Rereading the text can illicit patterns beyond just the words being said (Brooks 2016).

RQ3: Mixed-method

To find out where and how this impacts students lives I want to follow the academic and non-academic tasks of a small number of students to find out where AAI / GenAI is confronting students with its actions. Similar to what Ferrari and Scher (2000) did concerning tasks and procrastination only focusing not on task completion but on whether tasks are affected by AAI / GenAI and how students feel about this.

Students in project education will be asked to write down three tasks that they intend to complete for five consecutive days and will be asked to score along different dimensions how AAI / GenAI is affecting their completion of the task and how students deal with restrictions laid upon by school policy or assignment rules.

The analysis would then consist of comparing tasks among these dimensions, to categorize the type of tasks, their importance to the subject and to compare tasks and scoring between subjects in which way they subvert or conform to the rules.

Before doing this I would like to interview a small group of students to determine which kind of tasks and dimensions would make the most sense and what kind of AAI or GenAI tools are relevant to ask about. For the questions and categories questions from Sense of Agency-scale (SoAS) developed by Tapal, Oren, and Eitam (2017) will be adapted.

Sampling and data collection

As mentioned above the initial interviews will be conducted among a group of students. To come with a representative view of students I will seek out students in different areas of expertise. Within our institute of the HAN there are students that follow programs for law, economics, teaching, medical professions, building engineering, informatics amongst others. And in different years. For the first interviews it will be difficult to distribute this evenly.

For the surveys I will go through student organizations for the different fields of expertise, which have closer relationships with students, to be able to distribute widely as well as to acquire a large number of respondents. Within the institute there are about 30.000 students that are working on a bachelor-diploma, so it would seem feasible to come with enough respondents even if only 1% of active students respond.

For the mixed-method analysis a reward system would make sense as students would be spending quite some time on their task cards similar to the research performed by Ferrari and Scher (2000).

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